

$$\begin{aligned}
 \boxed{\hat{R}_k(\theta)} &\iff \begin{aligned}
 \hat{R}_x(\theta) &= e^{-i\hat{\sigma}_x\theta/2} = \cos(\theta/2) (|0\rangle\langle 0| + |1\rangle\langle 1|) - i\sin(\theta/2) (|1\rangle\langle 0| + |0\rangle\langle 1|) \\
 \hat{R}_y(\theta) &= e^{-i\hat{\sigma}_y\theta/2} = \cos(\theta/2) (|0\rangle\langle 0| + |1\rangle\langle 1|) + \sin(\theta/2) (|1\rangle\langle 0| - |0\rangle\langle 1|) \\
 \hat{R}_z(\theta) &= e^{-i\hat{\sigma}_z\theta/2} = e^{-i\theta/2} |0\rangle\langle 0| + e^{i\theta/2} |1\rangle\langle 1|
 \end{aligned}
 \end{aligned}$$